AI-Powered Full Stack Developer Program

ANSWER KEYS · INSTRUCTOR ONLY

Week 2 · Day 1

How the Web Works + HTML Core

Every bug-hunt's BUGS.md, the other answer files, and the HW3 answer key

Momo House · TSI Canteen · Aarav · Sunita · Didi

Stage 1, AI as Tutor: AI may explain; you type every line.

Tech Skills Institute (TSI) Nepal · Kathmandu

D1-01-see-the-letters-arrive · sample-findings.md

D1-01 · Sample findings (instructor)

File: classwork/D1-01-see-the-letters-arrive/solution/sample-findings.md

1. Ping

Offline: `ping localhost`. With Wi-Fi: `ping google.com`.

```
Reply from ::1: time<1ms
Reply from 142.250.194.78: bytes=32 time=38ms TTL=115
```

- `localhost` answers in under 1 ms: the "letter" never leaves your laptop.
- `google.com` answers in about 20–80 ms from Kathmandu. **DNS** (the phone book) turned the name into an IP address first: look at the first line, `Pinging google.com [142.250.194.78]`.

2. The Network tab

Open `index.html` with Live Server → F12 → **Network** → press F5.

7 parcels: 1 HTML + 1 CSS + 5 SVG.

Name	Type	Size (about)
index.html	document	1.4 kB (Live Server adds a small script, so it is a little bigger than the file)
style.css	stylesheet	0.4 kB
logo-momo-house.svg	svg+xml	2–4 kB
momo.svg	svg+xml	2–4 kB
momo-plate.svg	svg+xml	2–4 kB
thamel.svg	svg+xml	5–10 kB
swayambhu.svg	svg+xml	3–6 kB

- The `ws` line (`websocket`) is **Live Server** listening for your saves. It is not part of the page. Don't count it.
- There is no `favicon.ico` request because the page sets `<link rel="icon" href="data:,">`.
- Exact sizes depend on the Chrome version. The **count** (7) is the point.

The biggest parcel is usually `thamel.svg` (it has the most shapes).

3. View Page Source

Right-click → **View Page Source** (Ctrl+U). The `<h1>` is on **line 13**: `<h1>Momo House: a page made of parcels</h1>`. Live Server's script is added near the end, before `</body>`, so the top lines match the file.

4. Wi-Fi bonus

Any news site shows 50–200 requests. Every picture, font, script and advert is one more parcel.

D1-04-broken-skeleton · BUGS.md

D1-04 Broken Skeleton — answer key (instructor)

File: classwork/D1-04-broken-skeleton/solution/BUGS.md

Line = the line in the **starter** file.

#	File	Line	Bug	Fix	Tag
1	index.html	1	`<!DOCTYPE html>` is missing (line 1 starts with `<html>`)	Add `<!DOCTYPE html>` as the very first line	D(HH) doctype-first
2	index.html	1	`<html>` has no `lang="en"`	Write `<html lang="en">`	D(HH) html-lang-require
3	index.html	5	`<titel>` typo instead of `<title>`	Write `<title>Aarav's Page</title>`	V + D(HH) title-require
4	index.html	8	`<h1>` is never closed, so the paragraphs under it are giant and bold	Add `</h1>` at the end of the line	V + D(HH) tag-pair
5	index.html	10	Wrong nesting: `</p>` comes before `</strong>`	Last opened, first closed: `<p>I live in <strong>Kathmandu</strong>, near Dillibazar.</p>`	D(HH) tag-pair / D(F12)
6	index.html	15	The comment ends with `->` instead of `-->`	Close it with `-->`	V

How you could see each one

1. HTMLHint: "Doctype must be declared first". Without it Chrome uses old "quirks mode".
2. HTMLHint: "An lang attribute must be present on <html> elements". Screen readers need the language.
3. The tab shows the file name, not "Aarav's Page", and the words "Aarav's Page" appear at the top of the page.
4. Everything until the next heading is inside the h1. HTMLHint: "Tag must be paired".
5. The page looks almost normal. HTMLHint flags the pair, and F12 → Elements shows the browser moved the text out of the paragraph.
6. The comment never closes, so the line and footer text ("Made by Aarav") and everything after it disappear.

D1-06-canteen-news-bughunt · BUGS.md

D1-06 Canteen News — answer key (instructor)

File: classwork/D1-06-canteen-news-bughunt/solution/BUGS.md

Line = the line in the **starter** file.

#	File	Line	Bug	Fix	Tag
1	index.html	11	Two `<h1>` elements on one page	Make "New on the menu" an `<h2>`	R
2	index.html	14	`<p>` written as a real tag, so it vanishes and the sentence breaks in two	Write `<p>` to print the characters < and >	V
3	index.html	12	`&nbps;` typo (letters swapped)	Write ` `	V
4	index.html	18	`<blockquote>` closed with `</quote>`	Close it with `</blockquote>`	V + D(HH) tag-pair
5	index.html	19	`<em>` opened and never closed	Add `</em>` after "do not"	V + D(HH) tag-pair

How you could see each one

1. No tool complains. The rule is: one h1 per page (the page title). Sections are h2.
2. The page shows "Wrap text in" and "tags." on two lines. The `<p>` itself is missing.
3. The page shows the text `&nbps;` literally.
4. The blockquote never ends, so everything after it is pushed in (indented).
5. Everything after "do not" is italic, even the headings. (The indent comes from the blockquote; the italic comes from this.)

D1-07-broken-lists · BUGS.md

D1-07 Broken Lists — answer key (instructor)

File: classwork/D1-07-broken-lists/solution/BUGS.md

Line = the line in the **starter** file.

#	File	Line	Bug	Fix	Tag
1	index.html	12	Three `<li>` with no `<ul>` around them	Wrap the three items in `<ul>` … `</ul>`	V + D(F12)
2	index.html	22	Cooking steps (order matters) are in a `<ul>`	Use `<ol>` … `</ol>` so the steps get numbers	V
3	index.html	19	`<il>` typo instead of `<li>`	Write `<li>Garlic</li>`	V + D(HH) tag-pair
4	index.html	31	The glossary uses `<li>` for the definitions instead of `<dd>`	Use `<dd>` … `</dd>` for each definition	V

How you could see each one

1. No indent and no bullets: the bullets hang off the left edge of the page. F12 → Elements shows the `<li>`s sit directly in `<body>`.
2. The steps show bullets, not 1, 2, 3, 4.
3. "Garlic" has no bullet. HTMLHint: "Tag must be paired".
4. The definitions show bullets and no indent. In a `<dl>`, each `<dt>` (word) is followed by a `<dd>` (meaning).

D1-09-broken-paths · BUGS.md

D1-09 Broken Paths — answer key (instructor)

File: classwork/D1-09-broken-paths/solution/BUGS.md

Line = the line in the **starter** file.

#	File	Line	Bug	Fix	Tag
1	index.html	9	Logo path uses `image/`, but the folder is `images/`	Write `src="images/logo-momo-house.svg"`	V
2	index.html	13	The About link points to `about.html`, but the file is in `pages/`	Write `href="pages/about.html"`	V
3	index.html	18	External link without `https://` becomes a relative path	Write `href="https://www.wikipedia.org"`	V
4	pages/about.html	9	Home link from inside `pages/` points to `index.html` (there is no index.html in `pages/`)	Go down one floor: `href="../index.html"`	V
5	pages/about.html	12	Image path from inside `pages/` has no `../`	Write `src="../images/thamel.svg"`	V
6	pages/menu.html	9	Home link is a path on Aarav's own laptop (`C:\Users\Aarav\…`)	Use a relative path: `href="../index.html"`	V

How you could see each one

1. A broken-image icon with the alt text "Momo House logo" shows instead of the logo.
2. Clicking "About us" gives a 404 / "Cannot GET /about.html".
3. Clicking "Wikipedia" looks for a file called `www.wikipedia.org` next to `index.html` → 404.
4. Clicking "Home" on the About page gives a 404.
5. The Thamel picture is broken on the About page.
6. Clicking "Home" on the Menu page does nothing or shows an error. It could only work on Aarav's laptop.

D1-11-broken-timetable · BUGS.md

D1-11 Broken Timetable — answer key (instructor)

File: classwork/D1-11-broken-timetable/solution/BUGS.md

Line = the line in the **starter** file.

#	File	Line	Bug	Fix	Tag
1	index.html	17	The header row uses `<td>` instead of `<th>`	Use `<th scope="col">` for the six header cells	V
2	index.html	19	The 11:00 row has only 4 day cells (it needs 5)	Add the missing cell: `<td>Lab</td>` for Thursday	V
3	index.html	20	`colspam` typo on the Lunch row	Write `colspan="5"`	V
4	index.html	22	"Project lab" has `rowspan="2"`, but the 2:00 row still has its own Thursday cell	Delete the last `<td>Lab</td>` from the 2:00 row (the rowspan already fills it)	V
5	index.html	23	Text "Tea break 3:00" is written between two rows, outside any cell	Make it a row: `<tr><td>3:00</td><td colspan="5">Tea break</td></tr>`	V

How you could see each one

1. The header row is not bold and not centred.
2. The 11:00 row is one cell short: there is an empty gap under Thu.
3. "Lunch" fills only the Sun column; the rest of the row is empty.
4. An extra cell sticks out to the right of the table on the 2:00 row.
5. The browser moves stray text out of the table: "Tea break 3:00" shows **above** the table.

D1-13-broken-signup-form · BUGS.md

D1-13 Broken Signup — answer key (instructor)

File: classwork/D1-13-broken-signup-form/solution/BUGS.md

Line = the line in the **starter** file.

#	File	Line	Bug	Fix	Tag
1	index.html	14	`<label for="fullname">`, but the input has `id="name"`	Make them match: `id="fullname"`	V
2	index.html	22	Phone uses `type="number"`	Use `type="tel"`	V
3	index.html	26	The two batch radios have different names (`batch1`, `batch2`)	Give both the same name: `name="batch"`	V
4	index.html	26	The radios have no `value`	Add `value="morning"` and `value="evening"`	D(URL)
5	index.html	33	`<textarea />` is self-closed	A textarea needs a real end tag: `<textarea …></textarea>`	V
6	index.html	36	The Sign up `<button>` is after `</form>`	Move the button inside the form, before `</form>`	V

How you could see each one

1. Click the words "Full name": the cursor does not jump into the box.
2. The box shows up/down arrows. Type 09841000000 and press the up arrow: it becomes 9841000001. The leading 0 is lost and the phone number is wrong.
3. You can tick Morning **and** Evening. Radios with one shared name let you pick only one.
4. Submit the form and read the URL: it says `batch=on`. The kitchen cannot tell which batch you chose.
5. The text box is full of code (`</p>`, `</form>` …) and the bottom of the page is gone. It swallowed the rest of the file.
6. Once the textarea is repaired, the button shows, but clicking it does nothing: it is not part of the form.

D1-15-momo-house-repair · BUGS.md

D1-15 Momo House repair — answer key (instructor)

File: classwork/D1-15-momo-house-repair/solution/BUGS.md

Line = the line in the **starter** file.

#	File	Line	Bug	Fix	Tag
1	index.html	12	Nav link `menu.htm` (missing the `l`)	Write `href="menu.html"`	V
2	index.html	17	The big momo picture has no `alt`	Add `alt="A steaming plate of momo"`	D(HH) alt-require
3	index.html	14	The last nav link `<a href="about.html">About` is never closed	Add `</a>` after "About"	V + D(HH) tag-pair
4	menu.html	29	`colspan="2"` in a 3-column table	Write `colspan="3"`	V
5	menu.html	32	"Today's specials" `<li>`s have no `<ul>`	Wrap them in `<ul>` … `</ul>`	V + D(F12)
6	menu.html	22	Heading jump: `<h1>` then `<h4>`	Use `<h2>Momo prices</h2>` (no skipped levels)	R
7	order.html	24	`<label for="mobile">`, but the input is `id="phone"`	Write `for="phone"`	V
8	order.html	41	`<textarea />` is self-closed	Write `<textarea …></textarea>`	V
9	about.html	17	`../images/chef-didi.svg`, but `images/` is next to the page	Write `src="images/chef-didi.svg"`	V
10	about.html	21	Instagram link without `https://`	Write `href="https://www.instagram.com/momohouse.ktm"`	V

How you could see each one

1. Click "Menu" on the Home page: 404.
2. HTMLHint: "An alt attribute must be present on <img> elements".
3. The heading and the text under it turn blue and underlined: they became part of the About link.
4. The achar row stops short: an empty gap with no border at the right end of the last row.
5. No indent and no bullets (they hang off the left edge). F12 → Elements shows `<li>` directly inside `<body>`.
6. No tool fires. The rule: headings go down one level at a time.
7. Click the word "Phone": the cursor does not jump into the box.
8. The notes box is full of code, and the Send button and footer are gone.
9. Didi's picture is broken. `../` goes out of the Momo House folder, where there is no `images/`.
10. Clicking it looks for a local file → 404.

D1-FF4-evil-page-bughunt · BUGS.md

D1-FF4 Evil page — answer key (instructor)

File: classwork/fast-finishers/D1-FF4-evil-page-bughunt/solution/BUGS.md

Line = the line in the **starter** file.

#	File	Line	Bug	Fix	Tag
1	index.html	11	`<img>` without `alt`	Add `alt="A plate of steamed momo"`	D(HH) alt-require
2	index.html	12	An `<a>` inside another `<a>`	Close the first link before the second one starts	D(F12)
3	index.html	14	A `<ul>` inside a `<p>`	End the paragraph before the list: `<p>Today we have:</p>`	D(F12)
4	index.html	21	The same `id="menu"` is used twice	Give the second heading its own id: `id="specials"`	D(HH) id-unique
5	index.html	22	`<li>` items inside a `<div>` (not a list)	Use `<ul>` … `</ul>` instead of the `<div>`	V + D(F12)
6	index.html	29	The number input has no `<label>`	Wrap the word: `<label for="plates">Plates</label>`	V
7	index.html	9	An `<h1>` inside a `<p>`	Remove the `<p>` and `</p>` around the heading	D(F12)
8	index.html	18	`&nbps;` typo	Write ` `	V

How you could see each one

1. The picture looks fine. HTMLHint underlines the tag.
2. F12 → Elements: the browser closed the first link early, so there are two separate links. Links can't nest.
3. F12 → Elements: the browser closes the `<p>` before the `<ul>` and makes an extra empty `<p></p>` from the stray `</p>`.
4. HTMLHint: "The id value [menu] must be unique". A link to `#menu` can only go to one place.
5. No bullets and no indent (the bullets hang off the left edge). F12 → Elements shows `<li>` inside a `<div>`, not a list.
6. Click the word "Plates": nothing happens. With a label, the cursor jumps into the box.
7. F12 → Elements: an empty `<p></p>` before and after the h1. A heading can't live inside a paragraph.
8. The page shows `Rs.&nbps;25` as text.

D1-FF5-notice-board-bughunt · BUGS.md

D1-FF5 Notice Board — answer key (instructor)

File: classwork/fast-finishers/D1-FF5-notice-board-bughunt/solution/BUGS.md

Line = the line in the **starter** file.

#	File	Line	Bug	Fix	Tag
1	index.html	11	A second `<head>` instead of `</head>`	Write the closing tag `</head>`	D(F12) + D(HH) tag-pair
2	index.html	45	A `<p>` written after `</html>`	Move the paragraph inside `<body>`, before `</body>`	D(F12) / D(W3C)
3	index.html	29	Heading jump: `<h2>` then `<h5>`	Use `<h2>` (one level at a time)	R
4	index.html	31	`<h4>` used only to make a sentence bold	Use `<p><strong>…</strong></p>`	R
5	index.html	27	`   ` used to make space	Use two separate `<p>` paragraphs	V
6	index.html	28	`</hr>` instead of `<hr>`	Write `<hr>` (a void tag: no closing tag)	V + D(HH) tag-pair
7	index.html	21	The nested list is after `</li>` instead of inside the `<li>`	Move `</li>` to after the inner `</ul>`	D(F12)
8	index.html	42	`<ol>` closed with `</ul>`	Close it with `</ol>`	D(HH) tag-pair
9	index.html	18	Bare text "Notices:" directly inside `<ul>`	Move it out: `<p>Notices:</p>` before the list	R / D(F12)
10	index.html	14	A space in the file name: `notice-board .svg`	Write `src="images/notice-board.svg"`	V
11	index.html	38	The link says `#events`, but the heading is `id="event"`	Make them match: `id="events"`	V
12	index.html	35	A row of `<td>`s directly in `<table>`, with no `<tr>`	Wrap the cells in `<tr>` … `</tr>`	D(F12)

How you could see each one

1. The page looks normal. The browser ignores the second `<head>` (F12 → Elements shows only one). HTMLHint also flags the unpaired tag.
2. It still shows! The browser moved it into `<body>` (F12 → Elements). Only the W3C validator complains.
3. No tool fires. The rule: never skip heading levels.
4. It is not a heading of a section. Headings are for the outline, not for bold text.
5. A big empty gap between two sentences. `<br>` is a pause inside one thought, not a spacer.
6. The divider line is missing. The browser ignores `</hr>`.
7. It looks almost the same. F12 → Elements shows the inner `<ul>` is a direct child of the outer `<ul>`, which is not allowed.
8. HTMLHint: "Tag must be paired". The list never ends, so the contact line under it is pushed in too.
9. A `<ul>` may only hold `<li>`s. The text shows, but it is not a list item. F12 shows it as a loose text node.
10. The notice-board picture is broken.
11. Click "Events" at the top (with a short window): nothing happens.
12. It looks right. F12 → Elements shows the browser invented a `<tr>` for you.

D1-HW1-kathmandu-treks-repair · BUGS.md

D1-HW1 Kathmandu Treks repair — answer key (instructor)

File: homework/_solutions/D1-HW1-kathmandu-treks-repair/BUGS.md

Line = the line in the **starter** file.

#	File	Line	Bug	Fix	Tag
1	index.html	6	`<title>` is never closed	Add `</title>`	V + D(HH) tag-pair
2	index.html	12	`Everest.svg` with a capital E, but the file is `everest.svg`	Write `src="images/everest.svg"`	H
3	index.html	9	Link `href="treks"` is missing `.html`	Write `href="treks.html"`	V
4	index.html	15	The nested `<ul>` is outside its parent `<li>`	Move `</li>` to after the inner `</ul>`	D(F12)
5	treks.html	17	The header row uses `<td>`	Use `<th>` for the header cells	V
6	treks.html	19	A leftover cell under the `rowspan="2"` season cell	Delete the extra season cell from the Gokyo Lakes row	V
7	treks.html	31	`#form`, but the form is `id="booking-form"`	Write `href="booking.html#booking-form"`	V
8	booking.html	25	Label `for="name"`, but the input is `id="fullname"`	Write `for="fullname"`	V
9	booking.html	30	Phone uses `type="number"`	Use `type="tel"`	V
10	booking.html	34	Start date uses `type="text"`	Use `type="date"`	V
11	booking.html	38	The trek radios have different names	Give all three `name="trek"`	V
12	booking.html	46	"Book now" is a `type="reset"` button	Use `type="submit"`	V

How you could see each one

1. The whole page is **blank**, and the tab shows a long line of code. The title swallowed the rest of the file.
2. On Windows the picture still shows. On a Linux web server (most of the internet) it is broken: file names are case-sensitive there.
3. Click "Our treks" on the Home page: 404.
4. It looks almost right. F12 → Elements shows the inner list is not inside "Everest region".
5. The header row is not bold.
6. An extra cell sticks out to the right on the Gokyo Lakes row.
7. The link opens the booking page at the top, not at the form.
8. Click "Full name": the cursor does not jump into the box.
9. Up/down arrows appear in the phone box.
10. No calendar picker. People type dates in many different ways.
11. You can tick all three treks.
12. Fill the form and click Book now: everything is wiped, nothing is sent.

D1-HW3 · Answer Key: How the Web Works + HTML Core

File: homework/_solutions/D1-HW3-Answer-Key.md

TSI Nepal · AI-Powered Full Stack Developer Program · Week 2, Day 1 · Instructor copy (share after students try)

Try every question before you read this. Say your answer out loud first; interviews are spoken.

Every code answer below was checked in Chrome (headless `--dump-dom`, October 2026), so the "what the browser built" lines are real, not guessed.

Part 1: Self-check answers

S1. DNS = the **phone book**: it turns a name (`momohouse.com.np`) into an address. IP address = the **precise home address** of the server computer (e.g. `142.250.193.14`). Response = the **parcel** the server mails back: the HTML, then the CSS and images, each as its own parcel.

S2.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Momo House</title>
</head>
<body>
  <h1>Momo House</h1>
</body>
</html>
```

Full marks need: DOCTYPE, `lang="en"`, `meta charset`, `meta viewport`, `title`, and a body. (The `<h1>` is a bonus.)

S3. `<head>` = **the label on the parcel**: information *about* the page (title, charset, viewport, links to CSS). You don't see it on the page. `<body>` = **what's inside the parcel**: everything the visitor sees (headings, text, images, links, tables, forms).

S4. The whole thing, `<a href="menu.html">Menu</a>`, is the **element**. `<a …>` and `</a>` are the opening and closing **tags**. `href` is the **attribute**. `"menu.html"` is its **value**. `Menu` is the content.

S5. `<p>`. `<` means "print a <, don't start a tag" (an escape code). See D1-06 bug 2.

S6. Shopping list → `<ul>` (order doesn't matter). Recipe steps → `<ol>` (order matters: you can't steam before you fold). See D1-07 bug 2.

S7. `<a href="../index.html">Home</a>`. `../` = "go downstairs one floor" (out of `pages/`). See D1-09 bug 4.

S8. `<img>` is a **void** tag: it has nothing inside it, so it has no closing tag. You stick it on the page, like a sticker; you don't put things in it, like an envelope. `alt` = **the description you read over the phone**: what a screen reader says, and what shows if the picture can't load.

S9. 2. The first column of the next row is already taken by the cell above (it spans 2 rows), so the next row needs only 3 − 1 = 2 cells. If it has 3, one cell sticks out on the right. See D1-11 bug 4.

S10. The radios have **different `name`s**, so the browser thinks they are two separate questions. Give them **one shared name** (`name="batch"`). With no `value`, the URL shows `batch=on`, so the kitchen can't tell which one was picked. Add `value="morning"` / `value="evening"`. See D1-13 bugs 3 and 4.

Part 2: Interview answer key

T1. What happens when you type `momohouse.com.np` and press Enter?

Say this: "It works like the postal system. One: the browser asks DNS, the phone book, for the address of `momohouse.com.np`, and gets an IP address. Two: it sends an HTTP request, a letter, to that address. Three: the server, like a warehouse, finds the HTML file and mails it back as a response, a parcel. Four: the browser reads the HTML and sees it needs more parcels, the CSS and the images, so it sends more requests. Five: it builds the page and paints it on screen. I can see every parcel in F12 → Network." **They're checking:** that you know a page is many request/response pairs, not one magic download. **Weak answer:** "The website opens." Or a 7-layer OSI lecture with no link to what the browser actually does. **Likely follow-up:** "What is DNS?" → "The phone book of the internet: it turns a name into an IP address."

T2. What is `<!DOCTYPE html>` for?

Say this: "It is the first line of the page. It tells the browser: this is a modern HTML5 page, use standards mode. Without it, the browser switches to 'quirks mode' and copies old, buggy behaviour from the 1990s, so some sizes and spacing come out wrong. HTMLHint flags it with `doctype-first`." **They're checking:** that you know it is an instruction to the browser, not an HTML tag. **Weak answer:** "It is a tag that starts HTML." (It is not an element and has no closing tag.) **Likely follow-up:** "Is it case-sensitive?" → "No, `<!doctype html>` also works, but we write it the standard way."

T3. Element vs tag vs attribute?

Say this: "Think of labeled boxes. The **tag** is the label: `<p>` opens the box and `</p>` closes it. The **element** is the whole box: opening tag, contents and closing tag together. An **attribute** is extra information written inside the opening tag, as `name="value"`, like `href="menu.html"` on a link or `alt` on an image." **They're checking:** precise vocabulary. Many juniors use "tag" for everything. **Weak answer:** "They are all the same thing." **Likely follow-up:** "Name a tag with no closing tag." → "`<img>`, `<br>`, `<hr>`, `<input>`, `<meta>`: void elements."

T4. `<strong>` vs `<b>`, `<em>` vs `<i>`?

Say this: "It's tone of voice. `<strong>` means 'this matters', and `<em>` means 'stress this word'. They carry meaning: a screen reader can say them with emphasis, and search engines notice them. `<b>` and `<i>` only change the look, bold or italic, with no meaning. So I use `<strong>` and `<em>` when the meaning matters, and I leave pure looks to CSS." **They're checking:** meaning (semantics) vs appearance. **Weak answer:** "Both make text bold." True for the look, but it misses the point. **Likely follow-up:** "Then when is `<b>` OK?" → "For text that should stand out with no extra importance, like a product name in a review. It's rare."

T5. Absolute vs relative path?

Say this: "Same house vs postal address. A relative path is like saying 'the kitchen': it starts from where the page is, e.g. `pages/about.html` or `../index.html`. An absolute URL is the full postal address with `https://`, e.g. `https://www.wikipedia.org`. `C:\Users\Aarav\Desktop\site\index.html` is a path on Aarav's own laptop. It works only on his computer. On the server or on my laptop it's a 404." **They're checking:** that you can predict where a link goes, which is the #1 broken-site bug. **Weak answer:** "Absolute is long, relative is short." **Likely follow-up:** "What happens with `href="www.wikipedia.org"`?" → "Without `https://` the browser treats it as a relative path, a file called `www.wikipedia.org` in my folder, so I get a 404 (D1-09 bug 3)."

T6. Why alt? When is `alt=""` right?

Say this: "`alt` is the description you read over the phone. A blind visitor's screen reader reads it, it shows if the image fails to load, and search engines use it. So 'Plate of 10 steamed chicken momos' is a good alt. If the image is only decoration, like a pattern strip, I write `alt=""` so the screen reader skips it. Leaving `alt` out completely is the bug: then the reader may read the file name." **They're checking:** accessibility awareness, which more and

more companies test. **Weak answer:** "So the code is valid." (That's a side effect, not the reason.) **Likely follow-up:** "What would you write for a logo?" → "The company name, e.g. `alt="Momo House"`."

T7. `name`, `id`, `for` in a form?

Say this: "`id` is the input's unique name tag on the page. `for` on a `<label>` points to that `id`, so clicking the label puts the cursor in the box. `name` is the column the kitchen files your answer under: it's what the server sees. When I submit with `method="get"`, the URL shows `name=value` pairs, like `?email=sunita%40mail.com`. No `name`, no data." **They're checking:** that you understand what actually gets sent. **Weak answer:** "`id` and `name` are the same." **Likely follow-up:** "How do you test it without a server?" → "Use `method="get"` and read the address bar after submitting."

T8. "The browser never shows an HTML error"?

Say this: "True: browsers are polite guessers. They never complain, they guess and carry on, so a mistake can look fine. To find mistakes I use: one, HTMLHint in VS Code with the Problems panel, `Ctrl+Shift+M`, my spell-checker for HTML; two, F12 → Elements, to see what the browser actually built; three, the W3C validator at validator.w3.org. For forms, I also submit and read the URL." **They're checking:** debugging habits. A junior who can find their own bugs saves the team time. **Weak answer:** "I look at the page carefully." **Likely follow-up:** "Give an example where Elements shows a hidden mistake." → "Three `<li>` with no `<ul>`: the text still shows, only the indent and bullets are gone, and Elements shows no list around them."

C1. Find three problems

Any three of these five:

1. No `<!DOCTYPE html>` on line 1 → quirks mode. **D(HH)** doctype-first.
2. `<html>` has no `lang="en"`. **D(HH)** html-lang-require.
3. `<titel>` is a typo for `<title>`. The tab shows the file path, and "Aarav's Page" appears **on the page**: Chrome moved the unknown `<titel>` element into the body. **V** + **D(HH)** title-require.
4. `<h1>` is never closed, so the paragraph becomes part of the huge heading. Chrome built `<h1>Hello Kathmandu <p>…</p></h1>`. **V** + **D(HH)** tag-pair.
5. Wrong nesting: `<strong>` opened inside `<p>` but closed after `</p>`. Fix: `<p>I live in <strong>Kathmandu</strong></p>` (last opened, first closed). Chrome repaired it silently. **D(HH)** / **D(F12)**.

Fixed version:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Aarav's Page</title>
</head>
<body>
  <h1>Hello Kathmandu</h1>
  <p>I live in <strong>Kathmandu</strong></p>
</body>
</html>
```

What earns full marks: naming *how* you'd see each one (tab, HTMLHint, Elements), not only the fix. (Same bugs as D1-04.)

C2. Predict the table

Output: 3 columns (header: Time, Sun, Mon; second row: 10:00 + one cell spanning 2 = 3). "Tea break" is **moved above the table**, outside it. **Why:** text is not allowed directly inside `<table>` between rows, so the browser moves the text out and puts it before the table. Chrome's DOM (checked) is `Tea break`

`<table><tbody><tr>…</tr><tr>…</tr></tbody></table>`. It also adds a `<tbody>` you never wrote. (Same bug as D1-11 bug 5.)

C3. Canteen form, reference solution

`canteen-form.html`

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>TSI Canteen order</title>
</head>
<body>
  <h1>TSI Canteen order</h1>
  <form method="get">
    <label for="email">Email</label>
    <input type="email" id="email" name="email" required>

    <label for="phone">Phone</label>
    <input type="tel" id="phone" name="phone" placeholder="98XXXXXXXX">

    <fieldset>
      <legend>Meal</legend>
      <input type="radio" id="veg" name="meal" value="veg">
      <label for="veg">Veg</label>
      <input type="radio" id="nonveg" name="meal" value="non-veg">
      <label for="nonveg">Non-veg</label>
    </fieldset>

    <button type="submit">Order</button>
  </form>
</body>
</html>
```

Tested with: html-validate (0 errors). Submitting `sunita@mail.com`, `9841000000`, Veg gives `?email=sunita%40mail.com&phone=9841000000&meal=veg`. **What earns full marks:** `type="tel"` (not `number`: a phone number is not a quantity; with `number` you cannot type `+` or a space, and the spinner arrow drops a leading 0: 09841000000 → ↑ → 9841000001), every input has a label with matching `for`/`id`, one shared radio `name` with a `value` on each, the button inside the form. `fieldset`/`legend` is a bonus. **Common ways candidates fail:** `type="number"` for the phone; radios with different names; labels without `for`; the button outside `</form>`.

C4. "I click Submit and nothing happens"

Two bugs:

1. `<textarea />` is **not** a void tag, so `/>` is ignored. The textarea stays open and swallows everything after it as text. Chrome (checked) put `</form>`, the button and `</body></html>` *inside* the textarea as plain text. So there is no button at all: you see the code inside the box. Fix: `<textarea id="msg" name="msg"></textarea>`. **Fix this one first; it hides bug 2.**
2. The `<button>` sits **after `</form>`**, so even when it shows, it is not part of the form and submits nothing. Move it inside the form.

Fixed:

```
<form method="get">
  <label for="msg">Message</label>
  <textarea id="msg" name="msg"></textarea>
  <button type="submit">Send</button>
</form>
```

They're checking: void vs non-void elements, and that one bug can hide another. (Same as D1-13 bugs 5 and 6.)

Talk-about-your-project prompts: what a good answer contains

- **Hardest bug in D1-15:** names the bug, the visibility tag (e.g. "the specials list had no `<ul>`; it looked fine, but F12 Elements showed no list: D(F12)"), and the tool used.
- **Catering form check:** "I submitted with `method="get"` and checked that every field appeared as `name=value` in the address bar; one was missing, so I added its `name`."

Remember the One Rule: you may only ship code you can explain, line by line.

D1-HW4-thamel-guesthouse-replay · BUGS.md

D1-HW4 Thamel Guesthouse replay — answer key (instructor)

File: homework/_solutions/D1-HW4-thamel-guesthouse-replay/BUGS.md

Line = the line in the **starter** file.

#	File	Line	Bug	Fix	Tag
1	index.html	1	`<!DOCTYPE html>` is missing	Add it as line 1	D(HH) doctype-first
2	index.html	7	`<title>` is inside `<body>`, not `<head>`	Move the `<title>` line up into `<head>`	D(F12)
3	index.html	12	`Room-Deluxe.svg` with capitals, but the file is `room-deluxe.svg`	Write `src="images/room-deluxe.svg"`	H
4	index.html	9	The guesthouse picture has no `alt`	Add a useful `alt`	D(HH) alt-require
5	booking.html	19	The Double row has no `<tr>`	Wrap the cells in `<tr>` … `</tr>`	D(F12)
6	booking.html	17	`<caption>` comes after the first row	Move `<caption>` to be the first thing inside `<table>`	D(W3C)
7	booking.html	30	Email uses `type="text"`	Use `type="email"`	V
8	booking.html	38	The Deluxe `<option>` is after `</select>`	Move it inside the `<select>`	V
9	booking.html	26	`requred` typo	Write `required`	V
10	booking.html	34	The room `<select>` has no `name`	Add `name="room"`	D(URL)
11	booking.html	42	The room-type radios have different names	Give both `name="room-type"`	V
12	booking.html	48	The Book now button is after `</form>`	Move the button inside the form	V

How you could see each one

1. The page looks fine. HTMLHint: "Doctype must be declared first".
2. The tab **still** shows "Thamel Guesthouse"! The browser is a polite guesser. F12 → Elements shows the title in the wrong box.
3. Works on Windows, broken on a Linux web server.
4. HTMLHint: "An alt attribute must be present on <img> elements".
5. It looks right. F12 → Elements shows the browser invented the `<tr>`.
6. The caption still shows on top. Only the W3C validator reports it: a caption must be the first child of the table.
7. Type `abc` as the email and submit: the form is sent. With `type="email"` the browser stops you.
8. Deluxe is missing from the drop-down, and the word "Deluxe" appears loose next to it.
9. Submit with an empty name: the form is sent anyway.
10. Submit and read the URL: there is no `room=` part. The kitchen (server) never hears which room.
11. You can tick Standard **and** With balcony.
12. Click Book now: nothing happens.